Vehicle Transfer Approval by OTP

Discussion note for the MIS team | trade.cmll.in | 01 November 2026

1. Purpose

Vehicle transfers between sites currently need the Project Manager to log in to the mobile app or website and press a button at two points in the chain. PMs are often in the field or on the phone, so transfers wait. This note explains how the module works today and proposes an OTP (one-time password) step so a PM can approve by reading a code to the MIS executive over a phone call, without logging in.

2. How the module works today

The module is Masters > Vehicle transfers. One transfer moves through four fixed stages. Each stage is restricted to the people assigned to that site, and every action is written to the audit trail.

Stage	Who acts	What happens
1. Submit	MIS User or MIS Manager at the source site	Selects the vehicle from Vehicle Master, destination site, transfer date, driver, diesel, KMR and HMR. Status: Awaiting source approval.
2. Release	Source-site Project Manager or Production Manager	Presses Release vehicle. Status: Awaiting destination MIS verification.
3. Verify	Destination-site MIS User or MIS Manager	Confirms the vehicle physically arrived. Status: Awaiting destination PM acceptance.
4. Accept	Destination-site Project Manager or Production Manager	Presses Accept vehicle. Status: Completed. Vehicle Master location changes to the destination site.

Other rules already in place

- One live transfer per vehicle. A second submission is rejected until the first one completes.
- Site scope is enforced. A PM can act only for sites assigned to them. Administrators can see every transfer but cannot approve.
- Each stage sends an in-app bell notification to the next approvers. WhatsApp is switched off for these messages today.
- Transfers synced from Oracle are historical and are treated as already completed.
- Every stage records who acted, when, and the full set of transfer fields in the audit log.

Stages 2 and 4 are the bottleneck. They are the only steps that need the PM personally logged in.

3. Proposed OTP approval

The system already has two WhatsApp OTP flows: password reset and log book deletion. The deletion flow is the right model. The code is sent to the approver's phone, the requester types it in, and the server checks it against a stored hash. The same approach applied to the Release stage looks like this. The Accept stage is identical with destination roles.

1. The source MIS user opens the pending transfer and presses Request release OTP.
2. The server generates a six-digit code, stores only its hash on the transfer record, and sends the code by WhatsApp to the registered mobile of the source-site PM. The message names the transfer number, vehicle, source, destination and date so the PM knows exactly what is being approved.

3. The PM reads the code to the MIS user over a phone call. No login is needed.
4. The MIS user types the code into the transfer screen. If it matches, the server performs the release exactly as if the PM had pressed the button, and records "Released by [PM name] via OTP, entered by [MIS name]".
5. The code expires after 10 minutes, allows 5 wrong attempts and one request per minute. These are the same limits used for deletion OTPs today.

The guided WhatsApp message

The message itself guides the PM, so no training is needed:

```
Caliber MIS vehicle transfer approval
Transfer VT-1234 | Dumper D-17
From Sasti OB to Durgapur, dated 03-11-2026
Your approval code is 482 915.
Share it only with the MIS executive who called you.
To refuse, reply NO or simply ignore this message.
```

4. What stays the same

- The existing Release and Accept buttons for a logged-in PM remain. OTP is an additional path, not a replacement.
- The four-stage chain, site scope rules, one-live-transfer rule and Vehicle Master update are unchanged.
- The audit trail gains a line for OTP approvals but otherwise records the same fields.

5. Technical outline

Area	Change
Transfer record	New fields: otpHash, otpStage, otpRequestedAt, otpAttempts, otpRequestedByLogin, otpApproverLogins.
API	Two new routes beside the existing ones: request OTP and verify OTP. Verify calls the existing release or accept logic, so audit and Vehicle Master handling are reused.
Approver phone	Taken from the PM's mobile number in Users & employees. If no number is registered, the OTP button is hidden and the normal login path remains.
WhatsApp	The approved authentication template can carry a bare code immediately. The richer guided message above needs a new utility template, which takes one to two days of Meta review.
Screen	Transfer card gets a Request OTP button, a six-digit entry field, a countdown and a resend link after one minute.
Security	Only the hash is stored. Codes are random, single use, time limited and attempt limited. Who requested and who entered the code are both recorded.

6. Decisions for the team

Question	Recommendation
Who may request and enter the OTP?	Only MIS Users or MIS Managers at the site whose PM is approving. Allowing anyone would weaken the control.
Site has two PMs?	Send the code to all assigned PMs. The first valid code entered wins. This matches how bell notifications fan out today.

Should the destination MIS verification (stage 3) also get OTP?	No. MIS users are already at a desk and logged in. Keep OTP for the two PM stages only.
Which WhatsApp message first?	Launch with the already approved bare-code template, then switch to the guided template once Meta approves it.
Expiry and limits	10 minutes, 5 attempts, 1 request per minute, same as deletion OTP.

7. Rollout

1. Agree the decisions in section 6.
2. Submit the guided WhatsApp template to Meta.
3. Build and test on the staging database with two test sites.
4. Pilot on one source and one destination site for two weeks.
5. Enable for all sites and brief the PMs with the one-line instruction: "Read the code only to the MIS person who called you."